



Version 1.0

Literature Review

August of 2026

1. Feedback & Evaluation

- **Proposed Scope:** The project covers a comprehensive list of 13 incidents, ranging from violent crimes to public emergencies.
- **Methodology:** Evaluation will focus on the effectiveness of using Spatial Temporal neural networks to transform passive surveillance into an active system.
- **Operational Feasibility:** The project identifies clear constraints, such as lighting and hardware requirements, which provides a realistic baseline for academic evaluation.
- **Feedback from instructor:** On review (Pending).

2. Suggested Improvements

- **Advanced Low Light Processing:** Implementing Generative Adversarial Networks (GANs) or Infrared imaging support to maintain high detection accuracy in non light conditions.
- **Multi Modal Alerting (Audio Integration):** While currently out of scope, future versions could integrate audio analysis to detect gunshots, screams, or glass breaking to complement visual data.
- **Mobile Response Ecosystem:** Developing a dedicated mobile application for field officers to receive real time alerts directly from the Aegis AI dashboard.
- **Biometric Correlation:** Future integration of person identification (once privacy protocols are met) to help authorities track suspects immediately after an incident is detected.
- **Expansion of Incident Types:** Increasing the detection categories from 13 to include "Natural Disasters" like floods or earthquakes in future updates.

3. Final Grading Criteria

- **Model Accuracy:** The ability of the Aegis AI model to correctly classify incidents with acceptable precision.
- **System Integration:** Successful link between the AI engine, the API, and the Centralized Dashboard.
- **Documentation Quality:** Quality of the Final Technical Report, including clear system architecture and project logs.

- **Implementation:** Successful training of the AI model and its ability to classify the 13 specified incident types.
- **Model Accuracy:** The ability of the Aegis AI model to correctly classify incidents with acceptable precision.
- **Testing:** Real time performance evaluation, measuring latency and accuracy across different video streams.